Module: integrations (LMS/CMS, GigaChat, Keycloak, storage)

English version of ../../modules/integrations.md.

1. Purpose and boundaries

External contours: exchange with LMS and CMS, an AI summary through GigaChat, sign-in through Keycloak, S3-compatible file storage, notifications. It does not: no integration is mandatory for start-up — without them the product works in a reduced but honest form.

2. Use cases

- Previewing a batch from LMS/CMS (dry run) before importing.
- Idempotent import by `contract_number`.
- An AI summary for a card; without a key a "service not configured" message is shown.
- Sign-in through Keycloak (OIDC + PKCE); local JWT remains the fallback.
- Failure: an external service is unavailable → an integration error, the core keeps working.

3. Contract

`/api/integration/*` (preview, import), `/api/auth/*` (local and OIDC sign-in). Internal adapters: `services/integration.py`, `services/gigachat.py`, `auth/keycloak.py`, `services/file_storage.py`.

4. Data

Owns the results of an import (entities are created through core models, not by direct inserts). Files live in S3/MinIO; only references are stored in the database.

5. Permissions

Import and preview — `admin` only. The summary — by role, with a restriction on the data passed along. Integration secrets are available to the backend only.

6. Dependencies

Outbound: the GigaChat API (TLS with the Ministry of Digital Development CA certificate in `backend/certs`), Keycloak, S3. Inbound: `api.*`.

7. Configuration

`GIGACHAT_CREDENTIALS/GIGACHAT_SCOPE`, `AUTH_MODE`, `KEYCLOAK_*`, `S3_*`, `MINIO_ROOT_USER/PASSWORD`, `NOTIFICATIONS_ENABLED`. Empty values mean the integration is off, not a broken start.

8. Build and debugging

```
python -m pytest tests/test_integration.py tests/test_keycloak_auth.py tests/
test_gigachat_summary.py -q
```

9. Tests

`test_integration.py` (dry run and idempotency by `contract_number`), `test_keycloak_auth.py` (the PKCE flow), `test_gigachat_summary.py` (the summary without a key), `test_security_headers.py`, `test_cors.py`.

10. Observability

Integration errors are logged without secrets; the audit records an import. Errors: `integration_unavailable`, `oidc_error`, `storage_error`.

11. Limitations and readiness

- The LMS/CMS exchange is implemented by a mock adapter: the contract is described, the real system is not connected.
- Keycloak is started by the base compose but optional for the product.
- Readiness: without GigaChat/Keycloak the product starts and passes the smoke test.